

1. **Experiment:** Confocal Imaging of Complex Uptake and Lysosomal Colocalization
2. **Time:** 2026.07.14-2026.07.30
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4. **Materials:** STAR-Trap macrophages, control THP-1-derived macrophages, phorbol 12-myristate 13-acetate (PMA), Puromycin, complete medium (RPMI-1640 supplemented with 10% FBS and 1% penicillin-streptomycin), BeyoFC™ PE anti-human IL-6 Antibody (Beyotime, Cat. No. AC0806), ice-cold PBS, 24-well glass-bottom confocal plate, Lyso-Tracker Red (Beyotime, Cat. No. C1046), DAPI staining solution (Beyotime, Cat. No. C1006)

5. Method:

(1) Cell preparation:

- ① THP-1 cells were differentiated into macrophages by treatment with 100 ng/mL PMA for 48 h, followed by resting in fresh complete medium for 24 h.
- ② For STAR-Trap macrophages, the cells were transduced with lentivirus carrying the S2 construct, which was generated by introducing the designed mutations into the A2B3C3 (TOP1) scaffold, and selected with Puromycin (1 µg/mL) for 5 days. Untransduced THP-1-derived macrophages from the same batch were used as the control.
- ③ Both groups were seeded into 24-well glass-bottom confocal plates at 1×10^5 cells per well in complete medium, and were cultured under identical conditions (37 °C, 5% CO₂). Cell attachment, distribution and morphology were examined before imaging.

(2) Preparation of the fluorescently labeled complex:

- ① The fluorescently labeled complex was prepared as described previously, using the AF488 labeling system established in the earlier experiments.
- ② A single batch of labeled complex was used for both groups. The working aliquots were kept protected from light throughout.
- ③ The fluorescent signal was generated using Beyotime's BeyoFC™ PE anti-human IL-6 Antibody (Cat. No. AC0806), which recognizes human IL-6; therefore, the fluorescent signal traces IL-6.

(3) Incubation with the fluorescent complex:

- ① The following experimental groups were set up, with triplicate dishes per group:

Group	Description
Control	Control THP-1-derived macrophages incubated with the fluorescently labeled IL-6/sIL-6Rα complex, to observe baseline uptake and intracellular distribution
STAR-Trap	STAR-Trap macrophages incubated with the same batch of fluorescently labeled IL-6/sIL-6Rα complex, to evaluate receptor-mediated uptake and intracellular localization

- ② The fluorescently labeled complex was added to both groups at a final concentration of 1 mg/mL and in the same volume, and the cells were incubated at 37 °C in 5% CO₂ for 1 h so that the complex could contact the cells and be taken up.

(4) Removal of unbound fluorescent material:

- ① After incubation, the complex-containing solution was removed and the cells were washed with ice-cold PBS to lower the free fluorescent background.
- ② Washing was performed gently to avoid cell detachment and to prevent the samples from drying out.
- ③ Washing removes unbound material but does not by itself exclude complex that remains bound to the cell surface.

(5) Lyso-Tracker Red staining:

- ① Lyso-Tracker Red working solution was prepared in complete medium at a final concentration of 50-200 nM (Beyotime, Cat. No. C1046), pre-warmed to 37 °C, and protected from light.
- ② The cells were incubated with the working solution at 37 °C for 5-60 min, protected from light.
- ③ The working solution was removed and the cells were washed with pre-warmed PBS. Lyso-Tracker

Red was applied to live cells, as its retention depends on the lysosomal pH gradient.

(6) Fixation and DAPI staining:

- ① The cells were then fixed and washed with PBS.
- ② DAPI staining solution was added and incubated at room temperature for 10 min, protected from light.
- ③ Excess DAPI solution was removed and the cells were washed with PBS two to three times. A small volume of PBS was left in the dish to keep the cells moist during imaging.

(7) Confocal image acquisition and analysis:

- ① Images were acquired on a laser scanning confocal microscope. The green complex tracer channel, the red Lyso-Tracker channel and the blue DAPI channel were acquired separately, and merged images were generated.
- ② Excitation and detection windows were set according to the actual dyes and instrument configuration. Spectral cross-talk between the green and red channels was checked before acquisition.
- ③ DAPI was used to locate the nuclei only, and does not by itself define the complete cell boundary.
- ④ Spatial overlap of the green and red signals indicates that the two signals are close to each other within the resolution of the microscope, but does not by itself demonstrate degradation.

6. Result:

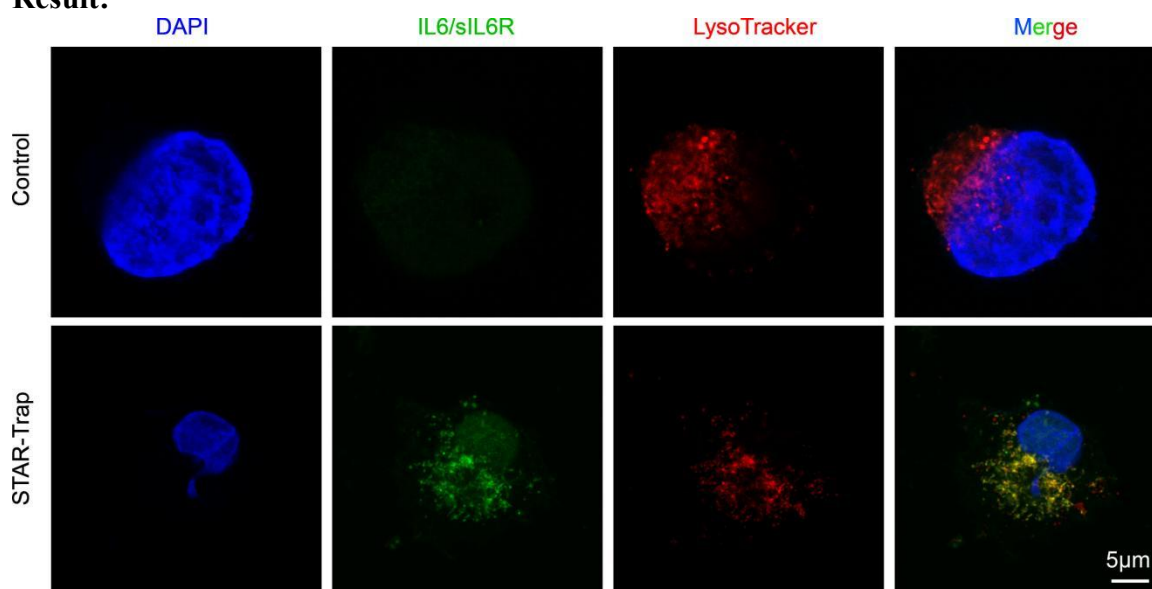


Figure 1. Confocal fluorescence images showing the intracellular distribution of the fluorescent IL-6/sIL-6R α complex probe and its colocalization with lysosomes in control and STAR-Trap. Nuclei were stained with DAPI (blue), the IL-6/sIL-6R α complex probe is shown in green, and lysosomes were labeled with LysoTracker (red). Scale bars, 5 μ m.

- (1) Uptake of the fluorescent complex: Confocal images showed clear punctate green fluorescence associated with the STAR-Trap cells, whereas almost no green fluorescence was detectable in the control cells. The control cells therefore showed little detectable complex-associated fluorescence, whereas STAR-Trap promoted the uptake of complex-related material.
- (2) Colocalization with acidic compartments: Part of the green signal in the STAR-Trap group overlapped spatially with Lyso-Tracker Red-positive acidic compartments, consistent with an association of the complex-related fluorescence with acidic endosomal or lysosomal compartments.
- (3) Interpretation: DAPI was used only to locate the nuclei and does not define the complete cell boundary. The green-red overlap indicates proximity within the resolution of the microscope; whether the complex-related material is degraded requires independent protein detection or degradation assays. The green signal traces the AF488-labelled sIL-6R α of the probe.